

Maths 141H – Homework 3

Due: **Beginning of class on 11/8/2024**

Rules

- This homework consists of **5 questions**, all carrying equal weight (although they are of unequal difficulty!). Please answer all questions. There is also an extra credit question at the end, worth an additional 10% (so it is possible to get up to 110%).
- Work in groups of **2-3 people**. Each group should submit **a single solution**, clearly indicating the group members (names and UIDs are required). **Groups must have been decided by 11/1** and announced to me by email, before the end of the working day. If you cannot find a group, let me know. **All members of the same group will receive the same grade.**
- Submit your answers at the beginning of class on 11/08/2024. Ensure your answers are legible (illegible parts will be ignored), and to help with the **markinggrading**,¹ please answer each question on a fresh page and staple everything together.
- **Good luck!**

¹A happy **markergrader** is a generous marker! Keep this in mind.

Question 1 – Integration and Sequences

1. Evaluate each of the following integrals:

(a) $\int \ln^2(x)dx.$

(b) $\int \cos(\ln(x))dx.$

(c) $\int x(x-1)^{100}dx.$ [*Hint.* I would not use the Binomial theorem.]

2. Evaluate each of the following integrals:

(a) $\int e^{\cos^{-1}(x)}dx.$ [*Hint.* First use the substitution $u = \cos^{-1}(x)$]

(b) $\int \tan^{-1}(\sqrt{x})dx.$

(c) $\int x^m \ln(x)dx,$ for $m \in \mathbb{N}_{>0}.$

3. Suppose that $\int f(x)dx = F(x).$ Find $\int f^{-1}(x)dx,$ in terms of $F(x).$

4. (a) Let $a_n = \sum_{i=1}^n \frac{i}{n^2}.$ Show that a_n is a Riemann sum for $\int_0^1 xdx,$ for each $n \in \mathbb{N}.$ Find $\lim_{n \rightarrow \infty} a_n.$

(b) Let $b_n = \sum_{i=1}^n \frac{i^2}{n^3}.$ Show that b_n is a Riemann sum for $\int_0^1 x^2dx,$ for each $n \in \mathbb{N}.$ Find $\lim_{n \rightarrow \infty} b_n.$

(c) Let $k \geq 1$ and $c_n = \sum_{i=1}^n \frac{i^k}{n^{k+1}}.$ Find a function $f(x)$ such that c_n is a Riemann sum for $\int_0^1 f(x)dx,$ for each $n \in \mathbb{N}.$ Find $\lim_{n \rightarrow \infty} c_n.$

Question 2 – Trigonometric Integrals/Substitutions

1. Evaluate:

$$\int \frac{\sin(x) - 5 \cos(x)}{\sin(x) + \cos(x)} dx.$$

2. (a) Show that the region enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

has area πab .

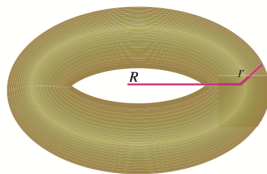
- (b) The Ellipse in Washington, D.C., which is located next to the White House, is approximately 1500 feet long and 1280 feet wide. Use the result of part 2 to determine how many square feet of grass should be purchased in order to sod the entire Ellipse.

[Hint. Assume that an equation of the Ellipse is $\frac{x^2}{750^2} + \frac{y^2}{640^2} = 1$.]

3. Evaluate:

$$\int_0^{\cosh(2)} \left(\frac{x-1}{x+1} \right)^{\frac{1}{2}} dx.$$

4. (a) In order to determine the prices of various sizes of doughnuts, a supermarket manager would like to have a formula for the volume of a doughnut:



The manager finds the formula:

$$V = \int_{R-r}^{R+r} 4\pi x \sqrt{r^2 - (x-R)^2} dx,$$

where $r < R$ in a calculus book but does not know how to evaluate the integral. Evaluate this integral using a trigonometric substitution.

- (b) Which doughnut should cost more, one with $(R, r) = (4, 2)$ or one with $(R, r) = (6, 1)$.

Question 3 – Partial Fractions

1. Evaluate the following integrals:

(a) $\int \frac{1}{x^4+1} dx$.

(b) $\int \frac{x^2-1}{x^3+3x+4} dx$.

(c) $\int \ln(x^2 + 1) dx$.

(d) $\int x \tan^{-1}(x) dx$.

2. Evaluate the following integrals, using the indicated substitution:

(a) $\int_{-\frac{5}{3}}^{-1} \sqrt{\frac{x+1}{x-1}} dx$, using the substitution $u = \sqrt{\frac{x+1}{x-1}}$.

(b) $\int_{-\frac{1}{9}}^{-\frac{1}{2}} \frac{1}{x} \left(\frac{x}{x+1}\right)^{\frac{1}{3}} dx$, using the substitution $\left(\frac{x}{x+1}\right)^{\frac{1}{3}}$.

3. Evaluate:

$$\int \frac{x^2}{(x^2 - 4)^2} dx.$$

Using this, and the substitution $u = \sqrt{x+4}$ evaluate:

$$\int \frac{\sqrt{x+4}}{x^2} dx.$$

4. Let $r \in \mathbb{Z} \setminus \{-1, 0\}$. Evaluate:

$$\int \frac{1}{x - x^{r+1}} dx,$$

by first using the substitution $u = x^r$. Using this, find:

$$\int \frac{1}{x - x^{11}} dx.$$

Question 4 – Numerical Integration

1. Prove that the Trapezoidal rule gives an exact value for $\int_a^b f(x)dx$ if f is a linear function.
2. Prove that Simpson's rule gives an exact answer for $\int_a^b f(x)dx$ if f is a polynomial of degree at most 3.
3. Let f be a function, $x_0 < x_1 < x_2$ real numbers in the domain of f and h a positive real number. Let $p(x)$ be the following polynomial:

$$p(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{h}(x - x_0) + \frac{f(x_0) - 2f(x_1) + f(x_2)}{2h^2}(x - x_0)(x - x_1).$$

Show that:

$$\int_{x_0}^{x_1} p(x)dx = \frac{h}{3} (f(x_0) + 4f(x_1) + f(x_2)).$$

4. Approximate the following integrals by using the Trapezoidal Rule and Simpson's Rule, using the indicated number of subintervals:
 - (a) $\int_1^3 \frac{1}{x} dx$, $n = 6$.
 - (b) $\int_0^1 e^{-x^2} dx$, $n = 10$.

Question 5 – Improper Integrals and Sequences

1. Show that

$$\int_0^\infty \frac{1}{x^n} dx$$

diverges for all $n \in \mathbb{Q}$.

[*Hint.* Consider the cases $n < 1$ and $n > 1$ separately.]

2. (a) Show that $\int_1^\infty \frac{1}{x^{\frac{3}{2}+1}} dx$ converges.

(b) Using the previous part, show that $\int_0^{\frac{1}{2}} \left(\frac{1}{x} - 1\right)^{\frac{2}{3}} dx$ converges.

3. Let f be a continuous function whose range is contained in its domain. Assume that $x_1 \in \text{dom}(f)$ and define $(x_n)_{n \in \mathbb{N}}$ as follows:

$$x_{n+1} = f(x_n), \text{ for } n > 1.$$

- (a) Suppose that $\lim_{n \rightarrow \infty} x_n = L$. Show that $f(L) = L$, i.e. that L is a fixpoint of f .
- (b) Let $f(x) = x^2 + \frac{3}{4}$ and $x_1 = 0$. Using the previous part show that the sequence $(x_n)_{n \in \mathbb{N}}$ diverges.
- (c) Let $f(x) = x^2 + \frac{1}{8}$. Assuming that $\lim_{n \rightarrow \infty} x_n$ exists, find it.
- (d) Let $a_1 = 2$ and $a_{n+1} = 2 - \frac{1}{a_n}$, for $n > 1$. Show that this sequence $(a_n)_{n \in \mathbb{N}}$ converges and find its limit.

[*Hint.* You may need to use a previous part of this exercise.]

4. Let $(a_n)_{n \in \mathbb{N}}$ be the sequence given by:

$$a_1 = \sqrt{2}, \text{ and } a_{n+1} = \sqrt{2 + a_n}, \text{ for } n > 1.$$

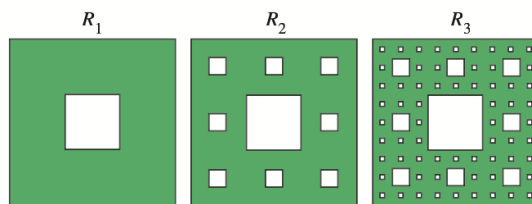
- (a) Show that $(a_n)_{n \in \mathbb{N}}$ is a bounded increasing sequence.
- (b) Show that $(a_n)_{n \in \mathbb{N}}$ converges.
- (c) Using the fact that:

$$a_{n+1}^2 = 2 + a_n$$

show that $\lim_{n \rightarrow \infty} a_n = 2$.

Question 6 – Extra Credit

- The **Sierpinski carpet** is constructed as follows: Begin with a square region R_0 of side length 1. Divide R_0 into 9 equal subsquares and remove the interior of the middle square. This gives us R_1 . For each of the remaining eight squares in R_1 perform the same operation. This gives us R_2 . We continue similarly:



The Sierpinski carpet S consists of all the points in R that are not removed in any of the steps, that is, of the intersection of R_n , for all $n \in \mathbb{N}$.

- Find the area of A_n of R_n , for any $n \in \mathbb{N}$.
 - Show that $\lim_{n \rightarrow \infty} A_n = 0$. What does this tell us about the area of the Sierpinski carpet. What do you think the perimeter of the Sierpinski carpet is?
- The **Fibonacci Sequence** is given by:

$$F_1 = 1, F_2 = 1, \text{ and } F_{n+1} = F_n + F_{n-1}, \text{ for } n > 2.$$

- Write down the first 10 terms of the Fibonacci sequence and show that this sequence diverges.
- Let $(G_n)_{n \in \mathbb{N}}$ be the sequence given by:

$$G_n = \frac{F_{n+1}}{F_n}, \text{ for } n \in \mathbb{N}.$$

Write down the first 10 terms of $(G_n)_{n \in \mathbb{N}}$ and show that:

$$G_n \in (0, 2), \text{ and } G_{n+1} = 1 + \frac{1}{G_n},$$

for all $n \in \mathbb{N}$.

- Show that $(G_n)_{n \in \mathbb{N}}$ is neither increasing nor decreasing. Using the previous parts, show that:

$$\lim_{n \rightarrow \infty} G_n = \frac{1 + \sqrt{5}}{2}.$$